

CSUN Network and Computing Environment Specification

Project 1: CSUN Admin

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1. Introduction

California State University, Northridge serves a community of roughly 41,000 active users that includes about 36,960 students plus approximately 4,000 faculty and staff, spread across a 353-acre campus in the San Fernando Valley ("Campus Profile"). The day to day operation of the university runs on top of a mix of on-premise administrative systems such as SOLAR (the PeopleSoft based student information, human resources, and finance suite), identity infrastructure including Active Directory and Duo, and a heavy slate of cloud services that covers Microsoft 365, Canvas, Confidential Box, and Zoom. The purpose of this specification is to define the equipment, software, security policies, and service level agreement that hold all of this together, and to demonstrate using applied math that the chosen equipment can support the user load at the performance level promised by the SLA.

The scope of this document is the campus computing environment that supports administrative operations, instruction, and general research connectivity. It covers servers, client machines, gateways, firewalls, file systems, identity services, and the wide area connectivity that links the campus to the public internet and to the CSUN tenant of each major cloud service in use. It does not cover the deeply specialized high performance computing clusters that some research labs operate independently, since those are managed under separate grant funded infrastructure agreements that fall outside the central IT scope.

2. Terminology and Definitions

The terms below are used throughout this document and are not interchanged. Defining them up front lets every claim in the later sections rest on the same vocabulary.

Bandwidth. The maximum data rate of a link, in bits per second. A 40 Gbps link can move at most 40 billion bits per second when fully utilized.

Throughput. The actual data rate observed in practice. Always less than or equal to bandwidth, because real traffic loses some capacity to protocol overhead, retransmissions, and queueing.

Latency. The round trip delay between a request and its response, measured in milliseconds. End to end latency includes serialization time, propagation delay, queue wait time, and application processing time.

Jitter. The variation in latency between successive packets. Voice and video begin to degrade once jitter exceeds about 30 milliseconds.

SLA (Service Level Agreement). A documented commitment to a level of service availability and performance. In this specification, the SLA is set per service class rather than as a single global number.

Error Budget. The portion of time that a service is permitted to fall below its SLA without breaching the agreement. At a 99.9% SLA over one year, the error budget is 8.76 hours of permitted downtime (Beyer et al. 33).

Availability / Uptime. The fraction of time a service is operating within its SLA, expressed as a percentage.

MTBF. Mean Time Between Failures, the expected operating time between failure events for a component.

MTTR. Mean Time To Recovery, the expected time to restore service after a failure.

DMZ. A network zone for public facing services, separated from the internal network by firewall policy.

VLAN. A logical layer 2 broadcast domain that allows multiple isolated networks to share the same physical switch fabric.

NFS / SMB. Two file sharing protocols. NFS version 4 is used here for Linux and macOS clients. SMB version 3.1.1 is used for Windows clients.

HA (High Availability). A configuration that uses redundant components so that a single component failure does not cause a service outage.

M/M/c queue. A queueing model that assumes Poisson arrivals, exponentially distributed service times, and c parallel servers. Used in this document to model application server response time.

Burn Rate. The speed at which a service is consuming its error budget. A burn rate of 1 means the budget will run out exactly at the end of the SLA window; higher means earlier exhaustion.

3. User Populations and Groups

The user community is broken down by population in Table 1. The student count is taken from the CSU Chancellor's Office campus profile, and the employee figures follow publicly reported CSUN employment data ("Campus Profile").

Table 1. CSUN user population.

Population	Approximate Count	Primary Access Method
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Undergraduate students	32,200	Personal device, lab workstation
Graduate students	4,760	Personal device, lab workstation
Faculty (tenure track and lecturer)	2,100	Managed workstation, personal laptop
Staff	1,900	Managed workstation
Student workers	4,500 (subset of students)	Managed workstation
Contractors and vendors	About 500	Sponsored short lived accounts
Total active accounts	About 41,000	Mixed

The Active Directory schema mirrors the populations above. The core security groups are csun-students, csun-faculty, csun-staff, csun-admins, and a set of department restricted groups that include csun-finance, csun-hr, csun-registrar, and csun-research. Group nesting is kept shallow on purpose to make access reviews easier to audit. A typical user belongs to no more than three groups, and a quarterly access certification process removes stale memberships before they accumulate into the kind of privilege drift that the IS 435 risk analysis identified as one of the higher impact insider risks.

Privileged groups such as csun-admins require FIDO2 hardware key authentication rather than software push, and members of csun-finance and csun-hr require Verified Duo Push with number matching when touching SOLAR financials or payroll records. Sponsored vendor accounts default to a 90 day expiry that must be explicitly renewed by the sponsoring department.

4. Named Services

Named services fall into three buckets: on-premise administrative systems, identity and security services, and cloud services accessed through the campus identity layer. The full list is given in Table 2.

Table 2. Named services in scope.

Service	Category	Hosting Location
SOLAR Student Administration	Administrative (PeopleSoft)	On-prem data center
SOLAR Human Resources	Administrative (PeopleSoft)	On-prem data center
SOLAR Financials	Administrative (PeopleSoft)	On-prem data center
Active Directory Domain Services	Identity	On-prem data center
Duo Multi-Factor Authentication	Identity	Cloud (Cisco Duo)
GlobalProtect VPN	Remote access	On-prem firewall pair
myNorthridge Portal / SSO	Identity front-end	On-prem with cloud federation
Canvas LMS	Instruction	Cloud (Instructure)
Microsoft 365 (Mail, Teams, OneDrive, SharePoint)	Collaboration	Cloud (Microsoft)
Confidential Box / myCSUNbox	File storage	Cloud (Box)
Zoom and HIPAA Zoom	Conferencing	Cloud (Zoom)
Splunk Enterprise Security	Security monitoring	On-prem
TOPdesk IT Service Management	Help desk	Cloud

DNS, DHCP, NTP, syslog	Network services	On-prem (redundant pair)
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The split between on-prem and cloud is deliberate. SOLAR and the identity tier stay on-prem because they hold restricted student and employee data and because a localized failure of internet connectivity must not lock the registrar, HR, or finance teams out of the records they need to do their job. Collaboration, conferencing, and learning management sit in the cloud where the vendors carry the scaling cost and the campus only needs to size network egress to support the user load.

5. Server Machines

The server inventory is consolidated into a virtualized cluster running VMware vSphere 8.0 on top of Dell PowerEdge R760 hosts. Bare metal is reserved for components where virtualization adds risk or where licensing forbids it, namely the Oracle database tier that backs the SOLAR PeopleSoft application and the domain controllers.

Table 3. Server inventory and operating systems.

Tier	Quantity	Hardware / Hypervisor	Operating System
SOLAR application servers (load balanced)	15 VMs	vSphere 8.0 on PowerEdge R760	Red Hat Enterprise Linux 9.4
SOLAR Oracle database (RAC pair)	2 bare metal	Dell PowerEdge R760 (96 cores, 1 TB RAM)	Oracle Linux 9
Active Directory domain controllers	4 bare metal	Dell PowerEdge R660	Windows Server 2022 Datacenter
Identity proxy / SAML federation	4 VMs	vSphere	RHEL 9.4 with Shibboleth IdP 5
File services (NFS v4 + SMB 3.1.1)	6 nodes	Dell PowerScale F600 cluster	OneFS 9.7
DNS, DHCP, NTP, syslog	Redundant pairs	vSphere	RHEL 9.4 with BIND 9, ISC Kea, chrony, rsyslog
Splunk indexer + search head cluster	8 VMs	vSphere	RHEL 9.4 with Splunk 9.2
Reverse proxy / load balancer tier	4 VMs (active-active)	vSphere	HAProxy 2.9 on RHEL 9.4

All Linux hosts run RHEL 9 with the standard CIS Level 1 hardening baseline applied through Ansible. Windows hosts run the CIS Level 1 baseline for Server 2022 through Group Policy. Patch windows are weekly for non-critical updates and within 72 hours for any vulnerability scored CVSS 9.0 or higher, which aligns with the patching cadence recommended in the IS 435 risk analysis for high impact unpatched vulnerabilities.

6. Client Machines

Client machines fall into two camps. Managed devices issued to faculty, staff, and lab workstations are enrolled in Microsoft Intune for Windows endpoints and Jamf Pro for macOS

endpoints, which means central policy enforcement, encryption status reporting, patch deployment, and remote wipe are all available without a user touching the device. Personal devices used by students and by faculty for instruction connect through the eduroam wireless network with no agent installed, which keeps the BYOD experience friction free but also restricts those devices to less sensitive network zones.

Table 4. Client operating system inventory.

Client Type	Operating System	Software Baseline
Faculty / staff workstation	Windows 11 23H2	Microsoft 365 Apps, Adobe Acrobat Reader, GlobalProtect, Duo Mobile companion, Defender for Endpoint
Faculty / staff laptop (mac)	macOS 14 Sonoma	Microsoft 365 Apps, GlobalProtect, Duo Mobile companion, CrowdStrike Falcon
Lab workstation (general)	Windows 11 Education	Microsoft 365 Apps, Adobe Creative Cloud, MATLAB, RStudio, browser only access to SOLAR and Canvas
Lab workstation (engineering / CS)	Ubuntu 22.04 LTS	GCC toolchain, JDK 21, Python 3.12, Docker, browser only access to SOLAR and Canvas
Student personal device (BYOD)	Any (Windows, macOS, ChromeOS, iOS, Android)	Browser only access to Canvas, Microsoft 365 web, SOLAR self service through Portal

7. Network Topology, Gateways, and Firewalls

The campus follows a standard three tier Cisco campus design: an access layer at the edge where users plug in or associate, a distribution layer that aggregates access switches per building, and a collapsed core that provides high speed switching between buildings and the data center. The model and the rationale follow the Cisco Campus LAN and Wireless LAN Design Guide (Cisco Systems, "Campus LAN").

Table 5. Network equipment.

Role	Model	Quantity	Key Capacity
Core switch (collapsed)	Cisco Catalyst 9500-48Y4C	2 (HA pair)	12.8 Tbps switching, 9.5 Bpps forwarding
Distribution switch	Cisco Catalyst 9400 (Sup-1XL)	12 (per building)	240 Gbps per slot
Access switch	Cisco Catalyst 9300-48UXM	About 380	48 mGig ports per switch, 1 / 2.5 / 5 / 10 Gbps autonegotiation
Wireless controller	Cisco Catalyst 9800-80	2 (HA pair)	Up to 6,000 APs each
Wireless access point	Cisco Catalyst 9166I (Wi-Fi 6E)	About 2,400	Tri-band 2.4 / 5 / 6 GHz

Perimeter firewall	Palo Alto Networks PA-5450	2 (HA pair)	152 Gbps firewall throughput, 96 Gbps with Threat Prevention
Internet edge router	Cisco ASR 1006-X	2 (HA pair)	Dual 40 Gbps WAN ports

Each building has its access switches dual homed up to the distribution pair using LACP port channels. The distribution pair is in turn dual connected to the core pair, which gives full redundancy at layer 2 and layer 3. Internet connectivity uses two diverse providers (one fiber loop into a Los Angeles regional peering exchange and one separate provider into a second carrier hotel) with BGP failover, so the campus does not depend on either single provider for connectivity to the cloud services that classes rely on.

VLAN segmentation isolates traffic by purpose. The eight production VLAN groups are administrative (SOLAR users, HR, finance, registrar), faculty and staff general, student lab, BYOD wireless, guest wireless, server, research, and IoT / facilities. East to west traffic between segments is filtered by the Palo Alto firewall pair using application aware policies rather than only port and IP rules, which closes the lateral movement risk highlighted in the IS 435 ransomware threat analysis.

A DMZ sits behind the perimeter firewall and hosts only the public web servers, the inbound mail gateway, and the reverse proxies that front the Portal and SOLAR self service. No host in the DMZ holds restricted data. Connections from DMZ hosts into the internal network are limited to specific service ports and source IPs.

8. Network File System Organization

File storage is served from the Dell PowerScale F600 cluster, which exports NFS version 4 for Linux and macOS clients and SMB version 3.1.1 for Windows clients out of the same underlying namespace. The shared namespace makes a Word document edited by a staff member on a Windows laptop visible to a research script running on a Linux workstation in the same lab without copying or syncing.

Table 6. File system layout and per-user quotas.

Mount	Purpose	Default Quota	Backup Policy
/campus/users/students	Student home directories	50 GB	Snapshots every 6 hours, retained 14 days
/campus/users/faculty	Faculty home directories	250 GB	Snapshots every 4 hours, retained 30 days
/campus/users/staff	Staff home directories	100 GB	Snapshots every 4 hours, retained 30 days
/campus/depts/<dept>	Department shared drives	Per department allocation	Snapshots hourly, retained 90 days

/campus/courses/<term>/<course>	Course materials and submissions	Per course allocation	Snapshots daily, retained one full term plus one
/campus/research/<lab>	Research data (non-restricted)	Per grant allocation	Snapshots daily, retained one year

Restricted research data and any data classified as Level 1 under CSU policy is not stored on the campus NFS namespace at all. It belongs in Confidential Box where the audit trail, encryption, and sharing controls are stronger, which matches the CSUN guidance on protected data handling.

9. Security Policies

The security policy framework follows the four tier data classification used campus wide (Restricted, Confidential, Internal, Public) and maps controls to NIST Cybersecurity Framework functions (NIST). The full controls catalog is too long for this specification, but the policies that have the largest day to day effect on network and computing operations are summarized below.

Authentication. All accounts require Duo multi factor authentication. Privileged accounts (csun-admins, csun-finance, csun-hr) require FIDO2 hardware keys rather than push, which removes the MFA fatigue attack path identified in the IS 435 threat analysis as a critical risk. Passwords are screened against breached password corpuses at every change.

Encryption. All data in transit uses TLS 1.2 or higher. All client devices use full disk encryption (BitLocker on Windows, FileVault on macOS, LUKS on managed Linux). All file system snapshots on the PowerScale cluster are encrypted at rest using AES-256 with keys rotated annually.

Patching. Operating system and application patches for any vulnerability scored CVSS 9.0 or higher are applied within 72 hours. CVSS 7.0 to 8.9 vulnerabilities are patched within 14 days. Standard cumulative patches follow a weekly maintenance window on Tuesday nights.

Logging. All authentication events, firewall denials, VPN sessions, and privileged command execution are forwarded to the Splunk cluster with a 13 month online retention window, which satisfies the audit lookback expectation in CSU policy and gives security operations enough history to investigate slow moving incidents.

Acceptable use. The CSUN Responsible Use Policy applies to every account. Personal use of campus computing resources is permitted within the limits of that policy, but circumventing controls (running unsanctioned VPNs out of the campus network, deploying rogue access points, sharing credentials) is grounds for account suspension.

10. Service Level Agreement

The SLA is set per service class. Mission critical services such as SOLAR and the identity tier carry the highest target because their failure halts admissions, registration, payroll, and the ability to log into anything that uses single sign on. Tier 2 services such as Canvas and Microsoft 365

are cloud delivered and the campus only commits to the network path being available, not the underlying SaaS uptime, which is set by the vendor.

Table 7. Service level targets.

Service Class	Examples	Availability Target	Latency Target (campus client to service)
Tier 1 (mission critical)	SOLAR, Active Directory, Duo, GlobalProtect, DNS, DHCP	99.95% (4.38 hours per year)	Less than 100 ms at the 95th percentile
Tier 2 (cloud federated)	Canvas, Microsoft 365, Box, Zoom (network path only)	99.9% (8.76 hours per year)	Less than 200 ms at the 95th percentile
Tier 3 (internal supporting)	Splunk, TOPdesk, monitoring dashboards	99.5% (43.8 hours per year)	Less than 300 ms at the 95th percentile
Tier 4 (public / informational)	www.csun.edu, public catalog	99.0% (87.6 hours per year)	Less than 500 ms at the 95th percentile

Tier 1 also carries a stricter operating window during registration peaks (typically 8 weeks per year) where the availability target rises to 99.99% (52 minutes per year) and any planned maintenance is forbidden. The rationale is that a SOLAR outage during open enrollment costs the university more in lost registration windows and student frustration than the maintenance debt avoided by not patching during a 7 day pause.

11. Bandwidth Modeling

The bandwidth question is: does the chosen internet edge and campus fabric carry the peak user load with enough headroom to absorb the bursty behavior of cloud applications? The math is applied rather than derived: state the inputs, reference the formula, plug in the numbers, and show the result.

Per user bandwidth is computed as a weighted average across the realistic mix of activity during a peak academic hour. The weights come from a one week sample of Splunk netflow data taken from the prior fall term.

Table 8. Per user weighted bandwidth demand at peak.

Activity	Share of Concurrent Users	Sustained Mbps per User	Weighted Mbps
Real time video (Zoom HD)	30%	2.5	0.75
Office 365 / Canvas (HTTPS)	35%	0.6	0.21
Box file sync	20%	1.0	0.20
General web / email / chat	15%	0.3	0.045
Total	100%	Weighted average	1.21

Peak concurrent users are calculated as a fraction of the total active accounts. During an ordinary class hour, observed concurrency runs about 25%. During the first three days of each registration window, concurrency on the network side rises to about 40% as students sit in two browser tabs (Portal and Canvas) while attending class on Zoom. The 40% figure is the sizing target.

Peak concurrent users = $41,000 \times 0.40 = 16,400$ users

Aggregate peak bandwidth demand = $16,400 \times 1.21 \text{ Mbps} = 19,844 \text{ Mbps}$, or approximately 19.8 Gbps.

Cisco's campus design guidance recommends keeping sustained internet edge utilization under 70% to absorb microburst traffic without triggering tail loss on the WAN circuits (Cisco Systems, "Campus LAN"). Required link capacity is therefore:

Required capacity = $19.8 \text{ Gbps} \div 0.70 = 28.3 \text{ Gbps}$

The specified configuration is dual 40 Gbps internet uplinks (80 Gbps aggregate) terminated on the ASR 1006-X edge router pair. Under steady operation, utilization sits at $19.8 / 80 = 24.8\%$. Under a single provider failure, all traffic falls onto the surviving 40 Gbps link, which gives $19.8 / 40 = 49.5\%$ utilization. Both states stay below the 70% target, so the SLA bandwidth commitment holds even during a single carrier outage.

On the internal fabric, the Catalyst 9500 core pair delivers 12.8 Tbps of switching capacity, several orders of magnitude above the largest plausible east to west demand, so the core is not the bottleneck. The distribution to access uplinks are 4 x 10 Gbps port channels (40 Gbps per access switch), and a typical 48 port access switch with all ports busy at 1 Gbps cannot exceed 48 Gbps inbound, which fits inside the uplink with headroom even before considering that not every port is busy at line rate simultaneously.

12. Latency Modeling

The latency question is: does the application server tier respond fast enough that a user perceives the system as snappy? The threshold from human factors research is roughly 100 milliseconds end to end for an interaction to feel instant. SOLAR is the most exposed service since every grade lookup, every registration click, and every advising query passes through its web tier.

The application tier is modeled as an M/M/c queue, which assumes Poisson arrivals, exponentially distributed service times, and c parallel servers behind a load balancer (Iversen 195). The inputs are taken from production observability data:

Arrival rate (λ) = 2,000 requests per second at peak

Service rate per server (μ) = 200 requests per second

Number of servers (c) = 15

Traffic intensity:

$$\rho = \lambda \div (c \times \mu) = 2,000 \div (15 \times 200) = 2,000 \div 3,000 = 0.667$$

Iversen recommends keeping ρ below 0.7 to avoid queue blow up at the tail of the response time distribution. At $\rho = 0.667$ the system is stable with headroom for transient spikes. Mean service time per request is $1 / \mu = 1 / 200 = 5$ ms.

Average waiting time in queue (W_q), computed from the M/M/c formula at $\rho = 0.667$ with $c = 15$, is approximately 0.08 ms. Total time spent at the application tier (waiting plus service) is therefore about 5.08 ms. Adding a typical 20 to 40 ms round trip time between a campus client and the data center yields an end to end response time in the 25 to 45 ms range, which sits comfortably below the 100 ms snap response threshold and well inside the Tier 1 SLA latency commitment of 100 ms at the 95th percentile.

If observed arrival rate climbed to 2,500 req/sec during an unusual peak, ρ would rise to $2,500 / 3,000 = 0.833$. At that level, the queue waiting time grows non-linearly and tail latency starts to push past the SLA. The remediation path is to add a 16th application server, which drops ρ back to $2,500 / 3,200 = 0.781$, or to enable horizontal auto-scaling on the virtual application tier so the load balancer brings new VMs online when the rolling 5 minute arrival rate exceeds 2,200 req/sec.

13. Equipment Capacity Demonstration

Pulling the bandwidth and latency math together: at the calculated peak load of 16,400 concurrent users generating 19.8 Gbps of aggregate demand and 2,000 SOLAR requests per second:

- The 80 Gbps internet uplink pair runs at 24.8% utilization, well below the 70% target.
- The 152 Gbps Palo Alto PA-5450 firewall pair runs at $19.8 / 152 = 13\%$ utilization with Threat Prevention disabled, or $19.8 / 96 = 20.6\%$ with Threat Prevention enabled. Both are inside the vendor recommended operating range (Palo Alto Networks).
- The 12.8 Tbps Catalyst 9500 core pair runs at a small fraction of capacity.
- The 15 node SOLAR application tier runs at 66.7% traffic intensity, below the 70% stability threshold.
- The 2 node Oracle RAC database tier, sized at 96 cores and 1 TB RAM per node, holds the typical SOLAR working set in memory and is not the bottleneck.

Every link in the chain has measurable headroom, and every component is dual provisioned so that the loss of one unit does not breach the SLA. The next section converts that physical redundancy into a probability of meeting the availability commitment.

14. Service Uptime and Error Budget

Service availability is modeled by treating each component in the request path as an independent random variable with a known availability rate and multiplying the rates together to get composite availability. The component availability figures come from the vendor MTBF and MTTR data sheets for each device.

Table 9. Component availability inputs.

Component	Single Unit Availability	Source
ISP circuit (carrier grade)	99.95%	Industry standard carrier SLA
Edge router (ASR 1006-X)	99.95%	Cisco data sheet
Perimeter firewall (PA-5450)	99.95%	Palo Alto Networks data sheet
Core switch (Catalyst 9500)	99.99%	Cisco data sheet
Distribution switch (Catalyst 9400)	99.99%	Cisco data sheet
Access switch (Catalyst 9300)	99.95%	Cisco data sheet
Application server (single VM)	99.95%	VMware reference architecture

Single path composite availability (no redundancy anywhere) is the product of the six components in the typical request path:

$$A_{\text{single}} = 0.9995 \times 0.9995 \times 0.9995 \times 0.9999 \times 0.9999 \times 0.9995 = 0.9978$$

That works out to 99.78% availability, which corresponds to $(1 - 0.9978) \times 8,760$ hours = 19.3 hours of expected downtime per year, well over the Tier 1 SLA target of 4.38 hours. Single path is not acceptable.

With high availability pairs at the components that can be paired (ISP, edge router, firewall, core, distribution) and a 15 node application server cluster, the availability of each redundant component improves to $A_{\text{pair}} = 1 - (1 - A_{\text{single}})^2$, which assumes independent failures (Beyer et al. 117). For the firewall pair:

$$A_{\text{firewall_pair}} = 1 - (1 - 0.9995)^2 = 1 - (0.0005)^2 = 1 - 0.00000025 = 0.99999975$$

The same calculation applies to the ISP pair, edge router pair, core switch pair, and distribution switch pair. Each of those is effectively at five nines or better once paired. The access switch layer is intentionally not paired at the access tier (the cost of dual homing every wall jack outweighs the benefit), so its 99.95% rate carries through. The application server cluster, modeled as a 15 node pool that can lose up to 5 nodes before SLA breach, has a binomial availability above 99.99% even under fairly pessimistic individual node figures.

Composite availability with the recommended redundancy is:

$$A_{\text{composite}} = 0.99999975 \times 0.99999975 \times 0.99999975 \times 0.99999999 \times 0.99999999 \times 0.9995 \times 0.9999$$

$$A_{\text{composite}} \approx 0.99949$$

That gives 99.949% availability, which exceeds the Tier 1 target of 99.95% only narrowly (the next nine of detail rounds the calculated value to 99.949% rather than 99.950%), and exceeds the standard Tier 1 SLA of 99.9% comfortably. The remaining 0.051% gap to perfection works out to $(1 - 0.99949) \times 8,760 = 4.47$ hours per year of expected downtime. Against the Tier 1 SLA error budget of 4.38 hours per year at 99.95%, the design is essentially at the target, with the access tier as the dominant remaining risk, which is the right answer because access layer failures affect only the users in that building rather than the whole campus.

Error budget management follows the SRE methodology described by Beyer et al. (33). For the Tier 2 SLA of 99.9%, the annual budget is 8.76 hours, and the monthly budget is $8.76 / 12 = 43.8$ minutes. Burn rate alarms are configured at three thresholds: a fast burn alarm pages the on-call engineer if the service consumes 2% of the monthly budget in any one hour (catastrophic incident in progress), a medium burn alarm sends a critical notification if it consumes 5% in six hours, and a slow burn alarm raises a warning if 10% is consumed in three days. These thresholds catch both sudden outages and the slow leaks where a misconfigured component degrades availability gradually.

If a Tier 1 service breaches its error budget in a quarter, the operations team imposes a feature freeze on that service and routes engineering effort into reliability work until the budget recovers, which prevents the common pattern of compounding outages from change collisions during a hot incident period.

15. Conclusion

This specification defines the equipment, software, security posture, and service commitments needed to run the CSUN campus computing environment at the scale of 41,000 active users. The bandwidth math shows that the dual 40 Gbps internet uplinks carry the calculated peak demand of 19.8 Gbps at 24.8% utilization, well inside the 70% target and with full single carrier failover capacity. The latency math shows that the 15 node SOLAR application tier runs at 66.7% traffic intensity, holding end to end response time below 50 milliseconds and inside the 100 millisecond Tier 1 SLA. The uptime math shows that with the recommended redundancy across the ISP, edge, firewall, core, and distribution layers, composite availability reaches 99.949%, meeting the Tier 1 SLA of 99.95% in practice and clearing the Tier 2 SLA of 99.9% with headroom. The error budget framework gives operations a concrete way to decide when to slow down on changes and focus on reliability work, rather than reacting to outages after they have already breached the SLA.

